

Explanations of Communities

Thanks for participating in this survey.

The questionnaire will be divided into three main sections:

1. Demographics
2. Introduction to Community
3. Quantitative and Qualitative Evaluation of Community Explanations

Please note that you have **20 mins** to complete this survey.

*** Indicates required question**

Demographics

1. **Full Name** *
- (As in Matric Card)

2. **Email Address** *

3. **Gender** *

⌵ Dropdown

Mark only one oval.

- ☐ Female
- ☐ Male
- ☐ Prefer not to say

4. **Age** *

⌵ Dropdown

Mark only one oval.

- ☐ 20 - 24
- ☐ 25 - 30
- ☐ 30 - 35
- ☐ > 35

5. **Familiarity with online social platforms** *

Mark only one oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Expert

6. **Familiarity with network analysis** (e.g., modularity, centrality, community detection, and community search) *

Mark only one oval.

	1	2	3	4	5	
Not	☐	☐	☐	☐	☐	Expert

7. **Is English your first language?** *

Mark only one oval.

☐ Yes

☐ No

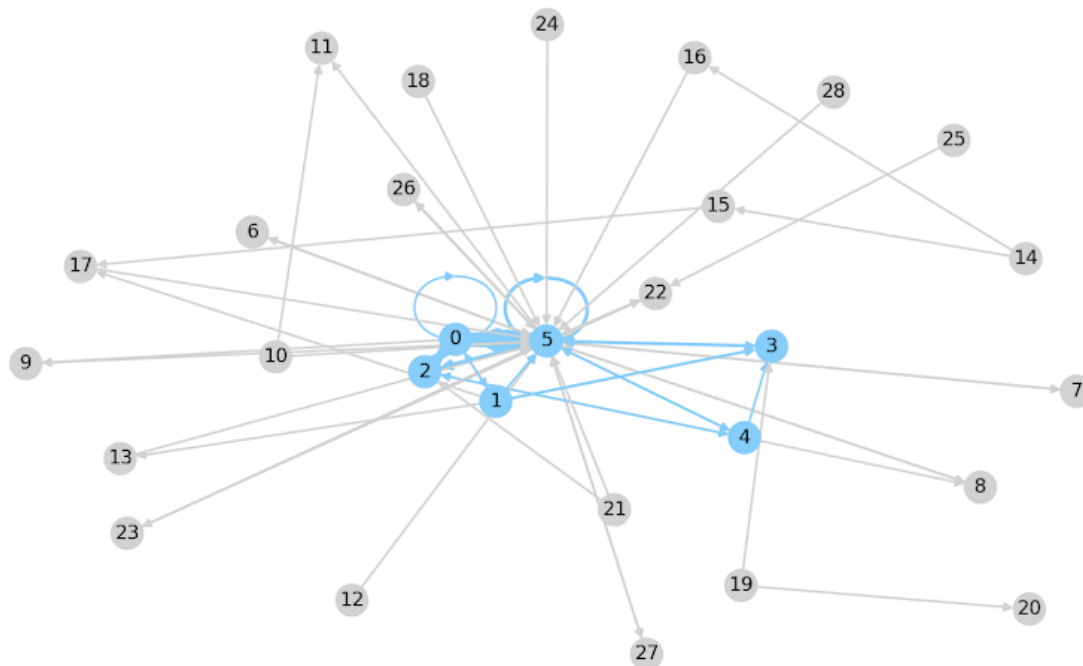
Introduction to Community

This community (consists of **6 users** and **157 interactions**) was extracted from an online social network, which was built from X (formerly Twitter) by modeling users' reply relationships into a network. In this network, users are represented as nodes connected by their reply interactions (tweets).

Simple Visualization of the community

In the figure below, the community within the online social network is highlighted in blue. Note that,

- + **Nodes:** represent users.
- + **Edges:** represent interactions between users:
 - A **self-loop** occurs when a user posts a tweet on their own.
 - An **edge** is created when one user replies to, or is replied to by, another user.
- + **Edge width:** indicates the total number of interactions between each user pair.
- + **Arrow direction:** shows whether the interactions are **one-way** or **mutual**.



Representative tweets for each user pair

(Marked with the total number of tweets between the pair)

User pair	Tweet (In total, 3 tweets)
1 → 3	Are you for real 🤔🤔🤔
1 → 3	🤔🤔🤔

User pair	Tweet (In total, 5 tweets)
3 → 5	You're so mad that Harry didn't marry you? It's sad!
3 → 5	What does this mean? 🤔🤔

User pair	Tweet (In total, 2 tweets)
0 → 1	As a human I can imagine Harry's sadness at his dad destroying his family with an affair and not having his back when his wife and child was being viciously attacked. I think hurting your child is a lot worse than children disappointing their parents.
1 → 0	I think you believe lies and don't care about proof. So many lies. They have to make money somehow.

User pair	Tweet (In total, 2 tweets)
4 → 2	Traitor he swore allegiance to the throne country and commonwealth
2 → 4	Agreed

User pair	Tweet (In total, 3 tweets)
4 → 5	Wouldn't fit in with the lies and M narrative Spoil the self pity feast
5 → 4	They can't keep there story straight

User pair	Tweet (In total, 45 tweets)
0 → 2	Common sense says u should watch the documentary so you can get both sides of the story. You can't make a good informed decision without knowing Harry's side.
2 → 0	I'm glad you didn't have the platform to destroy them to the world. With great privilege comes great responsibility. I'm sorry your family did you wrong. Mine has, too. But I still love them and forgive them.

User pair	Tweet (In total, 10 tweets)
2 → 5	Harry took \$100,000,000 to betray his father, grandmother and brother. What a good guy.
5 → 2	Family is everything 🤔
2 → 5	Wow. I had no idea. This is quite a list. 1985 was her first affair. Oh well.
5 → 2	And she confirmed it in interviews

User pair	Tweet (In total, 80 tweets)
0 → 5	I have no idea what you are speaking on in this tweet.
5 → 0	So you have not watch the documentary then lol #MeghanMarkle used #PrincessCatherine vedios in her #Netflix show
0 → 5	I don't hate William, I just dislike him and have no respect for him, especially being the eldest of my family.
5 → 0	Lol you don't want accepting the truth
0 → 5	How am I lying by just stating that I don't believe u watched the documentary and that you should watch it so you can get Harry's side?
5 → 0	I watched the documentary and it full of lies even American media had enough of there lies

User pair	Tweet (In total, 5 tweets)
5 → 5	Part 2 and there's more lol #Meghan #PrinceHarryisLiar
5 → 5	Even American media had enough of her lies
5 → 5	People refuse to wake up

Task Description

Now, you will be shown **FOUR** explanations of the extraction of this community.

Please **RATE** each explanation on its **explainability**, that is, how logical and well-founded an explanation is, and then provide your **justification**.

Explanations

Explanation 1:

Central node 3, threshold radius 1

Explanation 2:

This community is **conversation-heavy and centralized**. Rather than broadcasting content to a wide audience (few retweets, few quotes), members are primarily engaged in **direct exchanges** with each other. The dynamic is shaped by a few central users (0, 5, 2) who anchor the conversations, while smaller participants chime in occasionally. Peaks in likes suggest that certain comments struck a chord, but overall the community thrives on dialogue, not content amplification.

Explanation 3:

Score: 0.9868, with range [0, 2.0]

The community shows moderate cohesion, reflecting its general orientation toward developing and maintaining social relationships.

--> From the individual level, users in this community generally experience moderate enjoyment from interacting with other members, their relationships are generally neutral. Compared to the enjoyment they received outside the community, they experience moderate enjoyment with community members.

--> From the group level, compared to post information alone, group members demonstrate very high interaction preference, but their interaction frequency is very low.

Explanation 4:

Node-level features:

Node 0: {'clustering_coefficient': 0.3333333333333333, 'triangle_participation': 0.3333333333333333, 'eigenvector_centrality': 0.40192911697131134, 'expansion': 0.5}

Node 1: {'clustering_coefficient': 0, 'triangle_participation': 0.0, 'eigenvector_centrality': 0.20832737911452326, 'expansion': 0.5}

Node 2: {'clustering_coefficient': 0.6666666666666666, 'triangle_participation': 0.5, 'eigenvector_centrality': 0.4324928882413257, 'expansion': 0.16666666666666666}

Node 3: {'clustering_coefficient': 0, 'triangle_participation': 0.16666666666666666, 'eigenvector_centrality': 0.26729708839475724, 'expansion': 0.3333333333333333}

Node 4: {'clustering_coefficient': 1.0, 'triangle_participation': 0.3333333333333333, 'eigenvector_centrality': 0.33707694151686707, 'expansion': 0.3333333333333333}

Node 5: {'clustering_coefficient': 0.3333333333333333, 'triangle_participation': 0.8333333333333334, 'eigenvector_centrality': 0.6503337540801122, 'expansion': 3.1666666666666665}

Node-pair level features:

Feature: Jaccard, Values: {'nodes_0_1': 0.0, 'nodes_0_2': 0.2, 'nodes_0_3': 0.6666666666666666, 'nodes_0_4': 0.6666666666666666, 'nodes_0_5': 0.16666666666666666, 'nodes_1_2': 0.25, 'nodes_1_3': 0.0, 'nodes_1_4': 0.0, 'nodes_1_5': 0.4, 'nodes_2_3': 0.25, 'nodes_2_4': 0.25, 'nodes_2_5': 0.4, 'nodes_3_4': 0.3333333333333333, 'nodes_3_5': 0.0, 'nodes_4_5': 0.2}

Feature: Cosine Similarity, Values: {'nodes_0_1': 0.0, 'nodes_0_2': 0.3333333333333333, 'nodes_0_3': 0.8164965809277259, 'nodes_0_4': 0.8164965809277259, 'nodes_0_5': 0.23570226039551587, 'nodes_1_2': 0.40824829046386296, 'nodes_1_3': 0.0, 'nodes_1_4': 0.0, 'nodes_1_5': 0.5773502691896258, 'nodes_2_3': 0.40824829046386296, 'nodes_2_4': 0.40824829046386296, 'nodes_2_5': 0.47140452079103173, 'nodes_3_4': 0.4999999999999999, 'nodes_3_5': 0.0, 'nodes_4_5': 0.2886751345948129}

8. **Q1.** Consider the explainability of each explanation. Rank the four explanations from the most to the least agreeable.

*

Mark only one oval per row.

	1st	2nd	3rd	4th
Explanation 1	☐	☐	☐	☐
Explanation 2	☐	☐	☐	☐
Explanation 3	☐	☐	☐	☐
Explanation 4	☐	☐	☐	☐

9. **Q2.** Based on your ranking in Q1, please explain the rationale behind your choices.

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